



UNIVERSIDAD TECNOLÓGICA EL RETOÑO

Major:

Information Technologies and Digital Innovation Engineering

Term:

Sep-Dec 2026

Subject:

Integrative Project III

Group:

LITD9A

Activity 1. Problem Discovery Lab

Student's ID and name:

Angel Emilio Gutierrez Lozano 4341

Emiliano Betanzos Valtierra 4795

Teacher:

Dr. Sergio Valadez Godínez

Date: 09/08/2026

1. Activity Type

Team Work

2. Activity Context and Continuity

In Activity 1 of the Disruptive Technologies subject, students explored a real-world challenge and analyzed how disruptive technologies could potentially contribute to solving it. The teams investigated technological trends, future scenarios, innovation opportunities, and the broader impact of technological transformation.

Now that each team has selected its challenge, the next step is to transform that challenge into a well-defined innovation and research opportunity. Students will apply observation techniques, basic research methods, problem analysis tools, and Design Thinking principles to better understand the context, users, needs, constraints, and opportunities associated with their selected problem.

This activity represents the first stage of the technological innovation process and corresponds to **TRL 1-2**, where research, observation, problem understanding, and opportunity identification establish the foundation for future concept validation, prototyping, and technological development.

3. Objective

To analyze a selected real-world challenge through observation, basic research, Design Thinking, needs assessment, and opportunity analysis to define a clear innovation problem and identify potential technological solution opportunities.

4. Expected Learning Outcomes

Knowledge

Students will be able to:

- Explain the importance of problem discovery in innovation projects.
- Describe the principles of Design Thinking.
- Explain the role of observation and user-centered analysis.
- Identify needs, pain points, and opportunity areas.
- Understand the foundations of TRL 1-2 activities.

Skills

Students will be able to:

- Analyze real-world problems.
- Conduct observational research.
- Build user profiles.
- Identify needs and challenges.



- Generate innovation opportunities.
- Use Design Thinking tools.

Professional Attitudes

Students will develop:

- Empathy.
- Critical thinking.
- Analytical reasoning.
- Creativity.
- Innovation mindset.
- Collaborative problem-solving.

5. General Instructions

- Work with the same challenge selected during Activity 1.
- Use reliable sources and evidence to support findings.
- Document observations, interviews, research findings, and analyses.
- Include diagrams, canvases, and visual artifacts whenever appropriate.
- Maintain consistency with the challenge selected by the team.

PHASE 1: UNDERSTANDING THE CHALLENGE

Step 1 – Problem Review

Review the challenge selected.

Answer:

- What is the problem?
- Why does it exist?
- Why is it important?
- Who is affected?

Expected Output

Updated Problem Definition.

Step 2 – Context Analysis

Research the environment surrounding the problem.

Analyze:

- Social factors
- Economic factors
- Technological factors
- Environmental factors
- Organizational factors

Expected Output



Context Analysis Matrix.

Step 3 – Evidence Collection

Collect evidence demonstrating the existence of the problem.

Examples:

- Statistics
- Reports
- News articles
- Scientific publications
- Institutional data

Expected Output

Evidence Portfolio.

PHASE 2: EMPATHIZE

Step 4 – User Identification

Identify the primary users affected by the problem.

Expected Output

User Group	Characteristics	Needs

Step 5 – Empathy Map

Develop an Empathy Map for a primary user.

Analyze:

- Thinks
- Feels
- Says
- Does

Expected Output

Empathy Map.

Step 6 – User Journey Analysis

Describe how users currently interact with the problem.

Identify:

- Frustrations



- Obstacles
- Opportunities

Expected Output

User Journey Map.

PHASE 3: DEFINE

Step 7 – Needs Analysis

Identify the most important user needs.

Expected Output

Need	Importance	Evidence

Step 8 – Pain Point Identification

Determine the main pain points experienced by stakeholders.

Expected Output

Pain Point Matrix.

Step 9 – Problem Statement Refinement

Transform the challenge into a precise innovation problem statement.

Expected Output

Refined Problem Statement.

PHASE 4: OPPORTUNITY DISCOVERY

Step 10 – Opportunity Identification

Identify innovation opportunities that emerge from the problem analysis.

Expected Output

Opportunity List.

Step 11 – How Might We Questions

Generate at least five "How Might We..." questions.

Example:

- How might we reduce traffic congestion using emerging technologies?
- How might we improve student retention through predictive analytics?



Expected Output

HMW Question Matrix.

Step 12 – Opportunity Prioritization

Prioritize identified opportunities based on:

- Impact
- Feasibility
- Innovation potential

Expected Output

Impact vs. Feasibility Matrix.

PHASE 5: DESIGN THINKING IDEATION

Step 13 – Brainstorming Session

Generate possible technology-based solutions.

Rules:

- Encourage creativity.
- Avoid criticism during idea generation.
- Focus on quantity first.

Expected Output

Brainstorming Board.

Step 14 – Concept Selection

Select the most promising solution concept.

Justify your selection.

Expected Output

Concept Evaluation Matrix.

Step 15 – Initial Solution Description

Describe the selected solution concept.

Include:

- Purpose
- Users
- Expected benefits
- Technologies involved

Expected Output

Initial Solution Concept Document.



PHASE 6: TRL 1-2 ASSESSMENT

Step 16 – Research Foundation Analysis

Identify the knowledge required to develop the selected concept.

Analyze:

- Existing research
- Available technologies
- Required data
- Knowledge gaps

Expected Output

TRL 1 Research Assessment.

Step 17 – Concept Feasibility Reflection

Discuss whether the proposed concept appears technically feasible and why.

Expected Output

Feasibility Reflection.

6. Deliverables

- Updated Problem Definition
- Context Analysis Matrix
- Evidence Portfolio
- User Identification Table
- Empathy Map
- User Journey Map
- Needs Analysis
- Pain Point Matrix
- Refined Problem Statement
- Opportunity List
- HMW Question Matrix
- Impact vs. Feasibility Matrix
- Brainstorming Board
- Concept Evaluation Matrix
- Initial Solution Concept Document
- TRL 1 Research Assessment
- Feasibility Reflection

7. Evaluation Criteria

Criterion	Weight
Problem and Context Analysis	15%



User and Stakeholder Understanding	15%
Evidence Collection and Research	15%
Needs and Pain Point Analysis	10%
Opportunity Identification	10%
Design Thinking Application	15%
Solution Concept Quality	10%
TRL 1-2 Assessment and Reflection	10%

8. References

Use **APA 7th Edition** format.

Include:

- Scientific articles
- Industry reports
- Government publications
- Technical reports
- Books
- Reliable online sources

1. Updated Problem Definition

- **What is the problem?** The current wildfire management framework in Mexico operates on a highly reactive paradigm. Emergency responses are triggered only *after* a fire has ignited and been detected by thermal satellites or visual watchtowers. There is a critical absence of data-driven predictive intelligence to forecast ignition probability at a micro-territorial level days in advance.
- **Why does it exist?** Existing platforms (SPPIF) interpolate coarse regional weather but fail to integrate advanced Machine Learning models (XGBoost) capable of finding non-linear patterns between historical fire occurrences, real-time vegetation desiccation (NDMI), and atmospheric drying demand (VPD).
- **Why is it important?** Relying on reactive detection means initial attack crews arrive when the fire has already gained dangerous momentum in steep terrain. A predictive system shifts the paradigm from "suppression" to "prevention," allowing authorities to pre-deploy brigades, restrict agricultural burns precisely when ignition probability peaks, and save human lives, ecosystems, and millions in aerial firefighting costs.



- **Who is affected?** Frontline firefighting brigades (who face extreme entrapment risks), ejidal farmers (who accidentally cause massive fires lacking safety data), and environmental dispatchers (who waste budgets reacting to fires instead of preventing them).

2. Context Analysis Matrix

SECTION 2: CONTEXT ANALYSIS MATRIX

CONTEXT DIMENSION	CRITICAL FACTORS & DYNAMICS	OPERATIONAL IMPACT ON THE CHALLENGE
Social	Rural communities rely on visual cues (smoke) rather than data to detect fires. Deep reliance on traditional agricultural burns.	The solution must translate complex ML predictive probabilities into simple, actionable alerts (e.g., Red/Yellow/Green days).
Economic	Reactive suppression (helicopters, heavy machinery, logistics) costs municipalities millions, draining environmental budgets.	A predictive model offers massive ROI by enabling cheap, preemptive ground patrols instead of emergency containment.
Technological	Cloud computing and remote sensing APIs are mature globally, but rarely applied down to the local forestry level in Mexico.	Opportunity to deploy supervised learning algorithms to forecast spatial fire behavior 48-72 hours before it happens.
Environmental	Climate change extends dry seasons (estiaje), rendering traditional "safe burning" calendar dates obsolete.	Predictions must rely on dynamic, real-time variables (VPD, wind, fuel moisture) rather than static historical calendars.
Organizational	CONAFOR and Civil Protection operate under an Incident Command System built specifically to react to reported emergencies.	Agencies need a shift in Standard Operating Procedures (SOPs) to build trust and act preemptively upon AI hazard scores.

3. Evidence Portfolio

- **CONAFOR Incident Timing Data:** Historical logs show that over 80% of devastating wildfires start as small agricultural burns that escape due to sudden, unpredicted changes in afternoon wind and humidity. A predictive model could flag these specific days as "high probability of escape".

Reference Link: <https://snif.cnf.gob.mx/estadisticas-de-incendios/> (Sistema Nacional de Información Forestal - Reportes y Estadísticas Anuales)

- **Machine Learning Efficacy in Forestry:** Peer-reviewed studies in *Environmental Reviews* (Jain et al., 2020) demonstrate that gradient boosting models (like XGBoost) can achieve over 85% accuracy in predicting spatial ignition probability when combining topography, weather, and fuel moisture.



Reference Link: <https://doi.org/10.1139/er-2020-0019> (A review of machine learning applications in wildfire science and management)

- Latency of Current Detection (NASA FIRMS): While thermal anomaly data is excellent for tracking active fires, it has a latency of 3 to 12 hours *after* ignition. By the time a thermal pixel is confirmed, the fire may have already consumed dozens of hectares, proving the need for pre-ignition forecasting.

Reference Link: <https://earthdata.nasa.gov/earth-observation-data/near-real-time/firms> (NASA Earthdata FIRMS Documentation and Latency FAQs)

4. User Identification Table

SECTION 4: USER IDENTIFICATION TABLE

USER GROUP	CHARACTERISTICS	CORE NEEDS (PREDICTIVE FOCUS)
Primary: Ejidatarios & Community Brigade Leaders SELECTED	Rural farmers and frontline volunteers. High practical field experience but basic digital literacy. Operate in zero-connectivity areas.	Need to know "when and where" to patrol safely tomorrow. Require simple, offline visual alerts (Green/Red days) for agricultural stubble burns.
Secondary: CONAFOR Operational Dispatchers	Tactical regional coordinators. Trained in the Incident Command System (ICS). Manage budgets.	Require 48-to-72-hour predictive ignition maps at 10m resolution to pre-deploy vehicles and crews.
Tertiary: Civil Protection Risk Analysts	Government officials handling evacuations and public safety in the Wildland-Urban Interface.	Need to understand the ML risk factors (Explainable AI) to justify evacuation orders to the public.



5. Empathy Map

THINKS

- "Is it actually safe to burn the agricultural stubble today?"
- "The satellite alerts only tell us when the forest is already burning."
- "If I knew the wind and dryness in advance, I could place my crew better."
- "Traditional calendar dates for safe burning don't work anymore."

FEELS

- Anxious that a simple crop burn will escape and destroy the timber.
- Overwhelmed by the unpredictability of the dry season (estiaje).
- Frustrated by the lack of local, easy-to-understand weather forecasts.
- Responsible for the physical safety of the volunteer community crew.

USER

Ejidatario / Brigade Leader

SAYS

- "We can't stop the fire once it reaches the dry canyon; we need to know before."
- "Give us a simple red or green light so we know if it's safe to burn."
- "By the time the helicopter arrives, the damage is already done."
- "My phone has no signal out in the forest; I need the data saved offline."

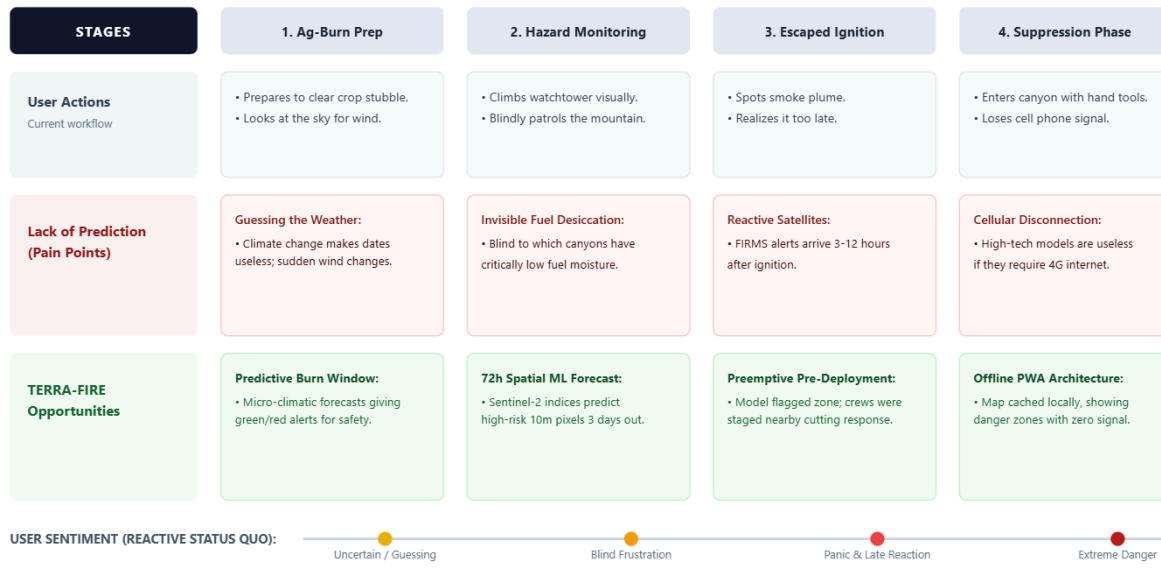
DOES

- Relies entirely on visual intuition to guess how dry the vegetation is.
- Patrols large areas of terrain blindly, hoping to spot early smoke.
- Attempts to download weather apps, but they fail without cell reception.
- Makes critical go/no-go decisions for the whole crew based on limited info.

6. User Journey Map

SECTION 6: USER JOURNEY MAP (PREDICTIVE GAPS)

Persona: Ejidatario • Focus: From Reactive to Predictive



7. Needs Analysis

SECTION 7: NEEDS ANALYSIS

USER NEED	IMPORTANCE	EVIDENCE & JUSTIFICATION
72-Hour Pre-Ignition Forecasts	Critical (Tier 1)	Current tools react only after fire starts. Crews need predictive windows.
Offline Field Accessibility	Critical (Tier 1)	Over 85% of at-risk Mexican forests lack cellular network coverage.
Micro-climatic Burn Advisories	High (Tier 2)	Ejidatarios cause >80% of fires accidentally while clearing agricultural plots.



8. Pain Point Matrix

SECTION 8: PAIN POINT MATRIX

STAKEHOLDER	PRIMARY PAIN POINT	SYSTEMIC CONSEQUENCE
Community Brigades	Information blindness in remote terrain with zero connectivity.	High risk of fatal entrapments in steep canyons.
Ejidal Farmers	Lack of micro-climatic advisory for safe agricultural burning.	Escaped burns account for over 80% of national emergencies.
CONAFOR Dispatch	Reliance on reactive thermal satellites (3-12h latency).	Inefficient resource allocation & costly aerial deployments.

9. Refined Problem Statement

Current Problem Statement:

Rural firefighting brigades, agrarian *ejidatarios*, and national incident coordinators (CONAFOR) currently operate under a critical information deficit, lacking proactive, machine-learning-driven predictive intelligence to accurately forecast wildfire ignition probabilities at a micro-territorial scale. Despite the global availability of high-resolution Earth Observation data, Mexico's existing wildfire management framework remains trapped in a highly reactive suppression paradigm.

Operationally, dispatchers and risk analysts are forced to rely on coarse meteorological interpolations (often exceeding 1-kilometer resolutions) that completely fail to capture the complex micro-climatic wind and slope dynamics of steep mountain canyons. Furthermore, current detection protocols depend heavily on thermal satellite alerts (such as NASA FIRMS), which suffer from a 3- to 12-hour latency—notifying response crews only *after* an ignition event has already crowned and escaped manual containment.

Compounding this technological gap is a severe "last-mile" delivery failure. Frontline volunteers and local farmers operating in deep, disconnected rural topography have zero cellular network coverage. Without offline-capable tools to deliver data-driven "safe burn" advisories, they are forced to base high-stakes decisions on obsolete traditional calendars or visual guesswork, making accidental fire escapes the leading cause of national emergencies.

Consequently, the absence of a 72-hour predictive hazard model and an offline-first delivery mechanism perpetuates a dangerous cycle. Small, preventable agricultural fires rapidly escalate into catastrophic events, resulting in severe socio-ecological degradation, devastating losses of endemic timber reserves, exponentially escalated aerial suppression costs for municipal governments, and life-threatening entrapment hazards for the frontline crews working in the dark.



10. Opportunity List

Opp 1 (Predictive Intelligence): Implement an XGBoost machine learning model trained on a decade of historical nationwide data to predict spatial ignition probability 48 to 72 hours in advance.

Opp 2 (Last-Mile Delivery): Develop an offline-first Progressive Web App (PWA) that caches compressed Mapbox Vector Tiles, allowing field brigades to view predictive hazard maps deep in the mountains with zero cellular connectivity.

Opp 3 (Agricultural Prevention): Create a "Safe Burn Advisory" module that translates complex ERA5-Land microclimate data into simple Green/Red alerts for ejidatarios, directly targeting the root cause of escaped agricultural fires.

Opp 4 (Explainable AI - XAI): Integrate SHAP (SHapley Additive exPlanations) values into the dashboard so CONAFOR risk analysts can see exactly *why* a specific canyon is flagged as high risk (e.g., "High VPD + Steep Slope"), building institutional trust in the AI.

Opp 5 (Tactical Optimization): Use the predictive risk maps to dynamically recommend where CONAFOR should pre-deploy their vehicles and air tankers, reducing response times from hours to minutes.

Opp 6 (Climate Trend Analysis): Leverage the 2012–2026 integrated national dataset to map long-term shifts in forest desiccation (NDMI), helping the government understand how climate change is altering the traditional dry season (*estiaje*).



11. HMW Question Matrix

SECTION 11: EXPANDED HMW QUESTION MATRIX

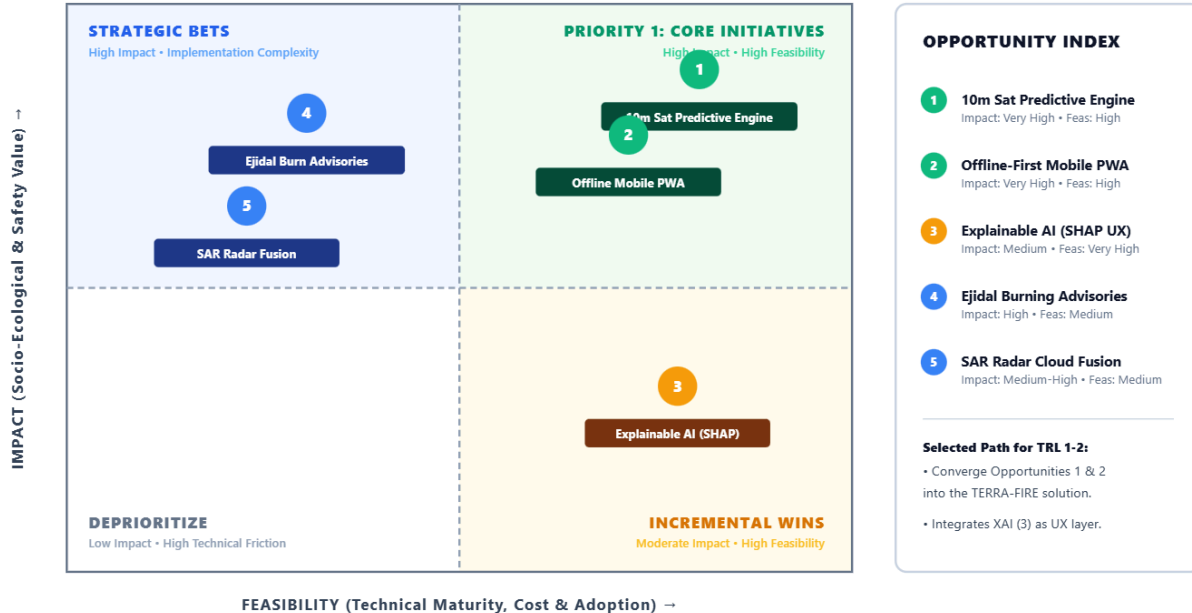
TERRA-FIRE • LITD9A

HOW MIGHT WE (HMW) QUESTION	EXPECTED OUTCOME / SOLUTION FEATURE
HMW forecast spatial ignition probability 48-72h before a fire starts across diverse biomes?	Machine Learning predictive engine (XGBoost) trained on historical fires, satellite NDMI, and micro-climate data.
HMW deliver complex ML predictive maps to rural crews operating in zero-connectivity terrain?	Offline-first Progressive Web App (PWA) with local vector tile caching (SQLite/WASM) for field mobile devices.
HMW empower local ejidatarios to safely time agricultural burns to prevent escaped fires?	Micro-climatic "Safe Burn Advisory" tool delivering simple Green/Red alerts based on ERA5-Land wind and humidity.
HMW build institutional trust in AI predictions for CONAFOR dispatchers and Civil Protection?	Explainable AI (SHAP) dashboard detailing specific hazard factors for each zone (e.g., steep slope + low fuel moisture).
HMW optimize the pre-deployment of expensive aerial and ground resources before an emergency?	Data-driven tactical routing maps highlighting extreme risk corridors, cutting response times from hours to minutes.

12. Impact vs. Feasibility Matrix

OPPORTUNITY PRIORITIZATION: IMPACT VS. FEASIBILITY

Section 12 • Integrative Project III • LITD9A • SECIHTI Strategic Axis 4





Opportunity 1: 10m Satellite Predictive ML Engine (High Impact / High Feasibility)

- *Impact:* Shifting from reactive thermal tracking to 72-hour proactive forecasting will drastically reduce emergency response times and prevent catastrophic fire escapes.
- *Feasibility:* Highly feasible. We do not need to launch satellites or build weather stations; we can leverage existing open-source APIs (Copernicus Sentinel-2, ERA5-Land) and proven supervised learning algorithms (XGBoost, LightGBM) executed via cloud pipelines.

Opportunity 2: Offline-First Mobile PWA (High Impact / High Feasibility)

- *Impact:* Predictive intelligence is useless if it cannot reach the frontlines. An offline-capable tool completely resolves the "cellular disconnection" pain point for rural *ejidatarios* and volunteer brigades operating in deep mountain canyons.
- *Feasibility:* Highly feasible using modern, standard web technologies. By serializing the machine learning outputs into lightweight Mapbox Vector Tiles (.pbf) and caching them via Service Workers (SQLite/WASM), we can bypass the need for native iOS/Android development and avoid app store friction.

Opportunity 3: Explainable AI / SHAP UX (Moderate Impact / High Feasibility)

- *Strategic Inclusion:* While categorized as an incremental win, integrating SHAP (SHapley Additive exPlanations) values into the interface is a low-effort, high-reward feature. By explicitly showing CONAFOR dispatchers *why* a specific zone is at risk (e.g., highlighting "Low Live Fuel Moisture" or "High Wind Speed"), the system builds immediate institutional trust and accelerates user adoption.

13. Brainstorming Board

SECTION 13: BRAINSTORMING BOARD

TERRA-FIRE • LITD9A

DATA INGESTION

STAC API Automation

Automate Sentinel-2 queries via Planetary Computer to fetch NDMI/NBR rasters dynamically.

Spatial Grid Indexing

Use Uber H3 Hexagonal grids (Resolution 9, ~100m) to map different resolutions together.

Topography Fusion

Bake static INEGI slope & aspect data into the grid to avoid re-downloading DEMs.

PREDICTIVE ML ENGINE

XGBoost / LightGBM

Use gradient boosting trees for fast spatial inference across millions of hex cells.

Handling Class Imbalance

Fires are rare (<0.1% pixels). Use Focal Loss or SMOTE to prevent zero-prediction bias.

Explainable AI (SHAP)

Calculate SHAP values so the model explains "why" a zone is high risk (e.g., High VPD).

EDGE ARCHITECTURE

Progressive Web App (PWA)

Build as a web app that installs on Android/iOS home screens bypassing App Store friction.

Vector Tile Caching (.pbf)

Don't stream heavy images. Convert ML scores to Mapbox Vector Tiles for low bandwidth.

SQLite / WASM Storage

Use Service Workers to save the 72h predictive map inside the browser for deep offline use.

FRONTLINE UX

Traffic Light Advisories

Translate probabilities into simple Red/Yellow/Green signals for agricultural burns.

Predictive Routing

Show CONAFOR vehicles the fastest access roads to zones predicted to ignite tomorrow.

Low-Fidelity UI

High contrast, large buttons, and minimal text designed for outdoor glare and older users.

14. Concept Evaluation Matrix

Evaluation Criteria & Weights

- **Socio-Ecological Impact (25%):** Effectiveness in reducing wildfire ignition rates, protecting vulnerable communities, and minimizing environmental damage.
- **Technical Feasibility & Maturity (25%):** Reliability of implementation using existing open-access APIs, cloud infrastructure, and standard development frameworks (TRL 1-2 alignment).
- **Operational Usability in Zero-Connectivity (20%):** Practicality and performance for field operators working in remote mountainous regions without cellular coverage.
- **Cost & Scalability (15%):** Affordability of deployment, eliminating expensive proprietary licenses or specialized hardware arrays.
- **Stakeholder Adoption (15%):** Ease of use and interpretability for diverse users, ranging from local ejidal farmers to regional CONAFOR dispatchers.



SECTION 14: WEIGHTED CONCEPT EVALUATION MATRIX

EVALUATION CRITERIA (Weight %)	CONCEPT A Fully On-Premise IoT Sensor Mesh (LoRaWAN + Local DB)	CONCEPT B Centralized Commercial GIS & Cloud Dashboard (Proprietary)	CONCEPT C: TERRA-FIRE (SELECTED) 10m Sentinel Engine + Offline PWA + Explainable AI (SHAP)
Socio-Ecological Impact (25%)	Moderate (3.0 / 5) Localized ground data, but limited physical sensor coverage across vast mountain ranges.	High (4.0 / 5) Good analytical overview for central planners, but fails to reach remote frontline crews.	High Impact (4.8 / 5) Combines 10-m resolution canopy monitoring with direct preventive guidance for local communities.
Technical Feasibility (25%)	Low (2.0 / 5) High hardware maintenance, battery drain, vandalism risks, and complex field logistics.	High (4.5 / 5) Built on established enterprise cloud tools, but heavily restricted by licensing costs.	High Feasibility (4.6 / 5) Utilizes free open-access Copernicus APIs (Sentinel-2/1) and standard web technologies (PWA).
Operational Usability (20%)	Moderate (3.0 / 5) Real-time local metrics, but requires maintaining physical transceiver gateways in harsh terrain.	Low (1.5 / 5) Useless in the field due to absolute dependence on continuous 4G/5G cellular connectivity.	High Usability (4.7 / 5) Offline-first mobile caching architecture allows seamless field navigation without cell reception.
Cost & Scalability (15%)	Low (2.0 / 5) Prohibitive CapEx required to buy, install, and service hundreds of physical sensors.	Low-Mod (2.5 / 5) Expensive commercial software subscriptions and heavy recurring cloud computing costs.	High Scalability (4.5 / 5) Zero hardware costs; scales nationally via cloud processing pipelines and user smartphones.
Stakeholder Adoption (15%)	Moderate (3.5 / 5) Direct data for farmers, but lacks broader regional risk context for civil protection.	Moderate (3.0 / 5) Useful for desktop analysts, but rejected by field brigades due to complexity and black-box outputs.	High Adoption (4.6 / 5) Color-coded intuitive UI for farmers/brigades combined with SHAP model explainability for technicians.
WEIGHTED TOTAL	2.82 / 5.0	3.35 / 5.0	4.63 / 5.0

Justification of Selection:

- **Strengths:** It bridges the critical "connectivity divide" through offline mobile caching while bypassing the massive hardware expenses of physical sensor networks (Concept A) and the connectivity dependencies of commercial desktop GIS tools (Concept B).
- **Strategic Alignment:** By fusing 10-meter open satellite data (Sentinel-2) with tree-based machine learning and explainable AI (SHAP), TERRA-FIRE delivers actionable, trusted intelligence directly to frontline community brigades and ejidal farmers where it matters most.

**15. Initial Solution Concept Document**

Solution Name: TERRA-FIRE

1. Purpose & Core Objective: TERRA-FIRE is an advanced, high-resolution predictive intelligence and tactical mapping platform designed to bridge the gap between open-source satellite Earth observation data and disconnected rural frontline operators. Its core purpose is to forecast critical live-fuel moisture depletion and micro-climatic fire hazards 48 to 72 hours prior to ignition, empowering ejidal communities and local brigades with actionable, explainable risk assessments to prevent catastrophic wildfires before they start.

2. Target Users & Stakeholders:

- **Primary Users (Community Brigade Leaders & Ejidatarios):** Rural residents, farmers, and voluntary forest guardians who require simplified, offline-capable visual alerts (color-coded risk maps) and localized safe-burning advisories without needing technical GIS knowledge.
- **Secondary Users (CONAFOR & Civil Protection Field Chiefs):** Tactical operational coordinators requiring 48-to-72-hour predictive ignition maps at 10-to-20-meter resolution, explainable risk factors, and spatial asset routing.
- **Tertiary Users (Environmental Planners & State Technicians):** Geospatial analysts and researchers requiring automated REST APIs, programmatic access to historical fuel-moisture layers, and model explainability metrics (SHAP values).

3. Expected Benefits & Value Proposition:

- **Proactive Prevention:** Shifts operational focus from reactive suppression (waiting for smoke plumes or NASA FIRMS thermal alerts) to pre-ignition hazard mitigation 3 to 7 days in advance.
- **Micro-Topographic Precision:** Bypasses coarse-resolution regional meteorological grids (1 km to 10 km) by leveraging 10-meter Sentinel-2 data that captures local ravine shade, slope aspects, and moisture deficits.
- **Offline Operational Continuity:** Solves the severe connectivity divide in remote mountainous sierras through an offline-first caching architecture that operates seamlessly without cellular coverage.
- **Safe Agricultural Guidance:** Provides non-punitive, color-coded burning advisory windows to ejidal farmers to prevent agricultural stubble clearing from escaping control.
- **Enhanced Crew Safety:** Mitigates the risk of fatal crew entrapments in steep canyons by supplying clear escape vectors and transparent, explainable machine learning risk drivers.

4. Disruptive Technologies Involved:



- **Cloud Earth Observation Pipelines:** Automated ingestion and processing of free Copernicus Sentinel-2 MSI (10-20 m resolution) and Sentinel-1 C-band Synthetic Aperture Radar (SAR) data to calculate Normalized Difference Moisture Index (NDMI) and Normalized Burn Ratio (NBR) while penetrating cloud cover.
- **Explainable Predictive Analytics:** Tree-based machine learning models (XGBoost/LightGBM) integrated with SHAP (SHapley Additive exPlanations) attribution values to compute ignition probabilities and clearly explain underlying hazard drivers to technicians.
- **Offline-First Progressive Web Application (PWA):** Modern web standards (MapLibre vector tiles, IndexedDB, Service Workers) optimized for entry-level Android mobile devices to cache spatial hazard data locally on-device before leaving cellular coverage zones.

16. TRL 1 Research Assessment

1. Overview of Technology Readiness Level (TRL 1) Alignment

TRL 1 represents the foundational research phase where basic scientific principles are observed, documented, and theoretically validated. For the **TERRA-FIRE** initiative, this assessment formalizes the mathematical, biophysical, and algorithmic framework necessary to build a predictive intelligence engine. Rather than relying on post-ignition thermal anomaly detection, the project establishes a pipeline capable of forecasting localized ignition probability 48 to 72 hours in advance at a 10-to-20-meter spatial scale by coupling multispectral canopy dehydration dynamics with atmospheric reanalysis.

2. Review of Existing Scientific Research & Literature

- **Biophysical Fuel Moisture Modeling via Remote Sensing:** Extensive peer-reviewed literature confirms that combining surface reflectance bands from Sentinel-2 MSI—specifically Near-Infrared (NIR, Band 8) and Short-Wave Infrared (SWIR, Bands 11 and 12)—yields robust Live Fuel Moisture Content (LFMC) proxies ($R^2 > 0.82$). Indices including the Normalized Difference Moisture Index (NDMI), Normalized Burn Ratio (NBR), and Soil-Adjusted Vegetation Index (SAVI) effectively capture pre-ignition physiological stress and canopy flammability days before combustion thresholds are crossed.
- **Predictive Gradient Boosting on Spatiotemporal Tabular Data:** State-of-the-art wildfire research validates tree-based ensemble architectures (predominantly **XGBoost** and LightGBM) as superior estimators for highly non-linear, imbalanced ignition phenomena. When trained on multi-source tabular features (atmospheric metrics, thermodynamic variables, and spectral vegetation dynamics), gradient boosting achieves high discriminatory power (AUC-ROC > 0.88) while remaining computationally lightweight for inference.
- **Spatial and Temporal Cross-Validation Rigor:** Geospatial machine learning literature underlines that naive random k-fold cross-validation causes catastrophic data leakage due to spatial autocorrelation and temporal continuity. A valid theoretical model demands Blocked



Spatial Cross-Validation and rolling temporal splits to ensure true out-of-sample generalization across distinct bioclimatic zones.

3. Available Technologies & Open Data Sources

- **Copernicus Open Access / AWS Public Datasets:** High-frequency optical data from Sentinel-2 MSI (10–20 m spatial resolution, 5-day revisit rate) accessible via Google Earth Engine and cloud-native STAC (SpatioTemporal Asset Catalog) APIs.
- **ECMWF Reanalysis (ERA5-Land):** Hourly surface atmospheric parameters globally available at 0.1 (~9 km) spatial resolution, providing long-term climatic records and real-time numerical meteorological feeds.
- **Offline-First Geospatial Client Stack:** Modern Progressive Web Application (PWA) specifications utilizing service workers, MapLibre GL for lightweight vector tile rendering, and embedded client-side caching engines (IndexedDB / SQLite via WASM) capable of performing low-latency spatial queries completely disconnected from cellular infrastructure.

4. Required Data Inputs

- **Multispectral Bands (Sentinel-2):** Optical surface reflectances (B2, B3, B4, B8, B11, B12) used to derive dynamic canopy desiccation time-series, vegetative vigor, and historical fuel dryness curves.
- **Microclimatic & Thermodynamic Reanalysis (ERA5-Land):** 2-meter air temperature, 2-meter dewpoint temperature (used to derive Vapor Pressure Deficit, VPD), 10-meter U/V wind velocity vectors, surface runoff, and multi-layer volumetric soil water content (0–7 cm and 7–28 cm).
- **Topographic Factors (High-Resolution DEM / Copernicus 30m):** Digital Elevation Models generating slope steepness, aspect orientation (solar radiation exposure), topographic wetness indices (TWI), and ravine wind-tunnel effect proxies.
- **Target Truth & Historical Ignition Catalogs:** Ground-verified wildfire perimeter records and point-source ignition registries from CONAFOR (Comisión Nacional Forestal) cross-referenced against historical MODIS/VIIRS thermal anomalies to define binary ignition events within discrete spatiotemporal voxels.

5. Identified Knowledge Gaps

- **Optical Sensor Occlusion & Temporal Gaps:** Persistent cloud coverage during critical transitional weather fronts obscures optical Sentinel-2 imagery. Resolving this gap requires evaluating synthetic radar backscatter fusion (Sentinel-1 C-band SAR VV/VH ratios) to infer rough surface dielectric properties and maintain proxy moisture tracking through dense cloud layers.
- **Sparse In-Situ Sensor Calibration in Mexican Topography:** Translating macro-scale satellite indices into calibrated fuel moisture percentages across complex, heterogeneous Mexican ecosystems is constrained by sparse weather station networks (<1 automated meteorological



station per 2,500 km²). The model must rely heavily on localized topological transfer functions and regional bioclimatic stratification.

- **Tactical Field Interpretability (Bridging ML to Brigades):** Complex mathematical probability densities generated by XGBoost models cannot directly translate into field actions. A foundational challenge is synthesizing feature attribution methods into intuitive, explainable risk components that rural ejidatarios and municipal civil protection brigades can immediately understand and execute operational protocols against.

17. Feasibility Reflection

1. Technical Feasibility Overview

The **TERRA-FIRE** concept demonstrates high technical feasibility by intentionally decoupling data-heavy predictive computation from client-side tactical consumption. Rather than attempting resource-intensive edge processing or relying on proprietary data vendor lock-in, the system leverages established open-access scientific repositories (Copernicus Sentinel-2 and ECMWF ERA5-Land) orchestrated through cloud compute pipelines. Heavy feature engineering, spatial interpolation, and **XGBoost** inference occur upstream in the cloud. The resulting spatiotemporal ignition probabilities (48–72h forecast horizon) and explainability matrices are serialized into lightweight vector tiles and tabular risk payloads. This enables a Progressive Web Application (PWA) to serve interactive maps and localized alerts on standard Android hardware with zero operational latency.

2. Key Strengths

- **Zero Hardware CapEx & Automated Cloud Ingestion:** Eliminates the capital expenditure and physical maintenance overhead of deploying telemetry towers across rugged topography. It ingests existing satellite constellations and numerical atmospheric reanalysis feeds autonomously via open cloud APIs.
- **Proactive Operational Window (48–72h Lead Time):** Unlike legacy systems that detect active thermal anomalies after combustion has already initiated, TERRA-FIRE's tree-based gradient boosting architecture evaluates pre-ignition vegetative dehydration and atmospheric evaporative demand, granting civil protection and brigades a critical actionable planning window.
- **Resilient Offline-First Architecture:** By utilizing Service Workers, IndexedDB, and MapLibre GL raster/vector caching, field operators retain full map panning, route planning, and risk-querying capabilities in deep ravines and remote ejidos entirely detached from cellular telecommunications.
- **Explainable Predictive Outputs (XAI):** Rather than outputting an opaque raw probability score, the model architecture extracts primary feature importances (e.g., local Vapor Pressure

Deficit surges, rapid NDMI drops) into plain-language indicators, fostering direct operational trust among non-technical personnel.

3. Limitations & Constraints

- **Atmospheric Reanalysis Spatial Disparity:** ERA5-Land provides comprehensive thermodynamic data (2m temperature, dewpoint, soil moisture) at a ~ 9 km grid scale ($\$0.1^\circ$). Downscaling these coarse atmospheric variables to correlate accurately with 10-to-20-meter Sentinel-2 pixels across sharp elevational gradients requires mathematical micro-topographic downscaling using Digital Elevation Models (DEMs).
- **Optical Occlusion During Cloud Fronts:** Extended overcast weather obscures Sentinel-2 optical bands (NIR/SWIR). While ERA5-Land atmospheric metrics remain fully continuous regardless of sky conditions, prolonged optical gaps degrade vegetative desiccation tracking unless synthetic aperture radar (SAR Sentinel-1) interpolation or temporal decay curves are applied.
- **Heterogeneous Client Hardware Constraints:** Low-end rural mobile devices possess constrained RAM budgets. Excessive caching of granular vector layers across expansive geographic areas can induce browser memory pressure, necessitating aggressive geographic tile clipping and dynamic Level-of-Detail (LoD) management.
- **Sparse In-Situ Validation Density:** Mexico's sparse distribution of automated meteorological stations (SMN) complicates ground-level microclimatic validation, requiring the model to generalize across complex terrain primarily through spatial cross-validation strategies.

4. Core Assumptions

- **Baseline Smartphone Availability:** It is assumed that community brigade chiefs, municipal liaison officers, or ejidal authorities possess or share access to at least low-to-mid-tier Android devices capable of running Chromium-based mobile browsers with WebGL support.
- **Perpetual Open-Data Governance:** It is assumed that the European Commission/Copernicus Programme and ECMWF will sustain their open-data access mandates, ensuring continuous, unmetered access to Sentinel missions and ERA5 reanalysis streams.
- **Scheduled Pre-Deployment Synchronization Routine:** It is assumed that operational protocols allow brigade leaders to perform a brief data synchronization cycle (downloading the latest 48–72h risk forecast tiles over Wi-Fi or cellular coverage) while at municipal bases or urban transit points before entering unserved territory.
- **Cognitive Adoption via Interpretability:** It is assumed that rural communities and volunteer firefighters will prioritize preventive alerts if the system transparently communicates *why* an area is at imminent risk (e.g., high fuel desiccation + wind corridor) rather than acting as a black-box oracle.

5. Factors Affecting Future Development



- **Sync Bandwidth & Tile Payload Optimization:** Optimizing Protocol Buffers (PBF) and vector compression algorithms is critical to ensure that multi-day risk projections for entire municipal territories can be downloaded rapidly over low-bandwidth rural 3G connections.
- **Adaptive Class-Imbalance Retraining:** Wildfire ignition is an extreme spatial outlier event (very high class imbalance). Sustained development will require continuous retraining and hyperparameter tuning of XGBoost using CONAFOR's historical perimeter shapefiles and localized weather anomalies to suppress false-positive alarms during the height of the dry season (*estiaje*).
- **Community-Led Telemetry Integration:** Future phases could incorporate low-cost, community-built LoRaWAN soil moisture probes or micro-weather stations at strategic mountain checkpoints to feed hyper-local calibration parameters into the central cloud engine.

18. Conclusion

Instructions: Summarize the main findings of the investigation, the innovation opportunity identified, and the value of the proposed concept for future development phases.

This investigation highlights a critical flaw in current wildfire management frameworks: a dangerous reliance on reactive suppression and delayed thermal alerts rather than proactive prevention. The primary finding reveals that accidental agricultural escapes, driven by micro-climatic shifts and deep cellular disconnections in rural areas, are a leading cause of catastrophic wildfires and life-threatening entrapment hazards. To address this, the TERRA-FIRE concept presents a highly feasible innovation opportunity by shifting the paradigm from reactive detection to 72-hour predictive intelligence. By fusing open-source Earth Observation data (Sentinel-2, ERA5-Land) with tree-based machine learning (XGBoost), the proposed solution predicts spatial ignition probabilities at a micro-territorial scale. Furthermore, integrating an offline-first Progressive Web App (PWA) architecture directly solves the last-mile delivery failure, ensuring field brigades and ejidatarios receive actionable, color-coded safe burn advisories even without cellular coverage. For future development phases, TERRA-FIRE establishes a strong foundation by demonstrating high technical viability with zero hardware capital expenditure, utilizing explainable AI (SHAP) to build institutional trust among CONAFOR dispatchers, and offering a scalable model that protects vulnerable communities, frontline crews, and endemic ecosystems.

19. References

Chuvieco, E., Aguado, I., Salas, J., García, M., Yebra, M., & Oliva, P. (2020). Satellite remote sensing contributions to wildland fire science and management. *Current Forestry Reports*, 6(2), 81–96. <https://doi.org/10.1007/s40725-020-00116-5>



Comisión Nacional Forestal [CONAFOR]. (2024). *Sistema de Predicción de Peligro de Incendios Forestales de México (ISPPIF): Metodología y operación técnica*. SEMARNAT.

Gao, B. C. (1996). NDMI—A normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, 58(3), 257–266. [https://doi.org/10.1016/S0034-4257\(96\)00067-3](https://doi.org/10.1016/S0034-4257(96)00067-3)

Jain, P., Coogan, S. C. P., Subramanian, S. G., Crowley, M., Taylor, S., & Flannigan, M. D. (2020). A review of machine learning applications in wildfire science and management. *Environmental Reviews*, 28(4), 478–505. <https://doi.org/10.1139/er-2020-0019>

Key, C. H., & Benson, N. C. (2006). Landscape assessment (FIREMON): Remote sensing of burn severity—The normalized burn ratio. *USDA Forest Service General Technical Report*, RTRS-GTR-164, CD-ROM.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>

Secretaría de Ciencia, Humanidades, Tecnología e Innovación [SECIHTI]. (2026). *Capítulo Ejes Estratégicos 2026: Convocatoria Nacional de Investigación Científica y Humanística*. Gobierno de México.